GRADE 3 · INTRO PYTHON TURTLE

Fix the Turtle Code

Ontario Math C3.2 — find and fix code

Name: ______________________ Date: ______________________

★ **I can find the bug in turtle code and fix it.**

Find it, then fix it

Sometimes turtle code draws the wrong thing. A coder reads it, decides what it SHOULD draw, and finds the wrong number.

A turtle-loop bug is usually the wrong range number (too few/many sides) or the wrong turn. Help Bit fix them.

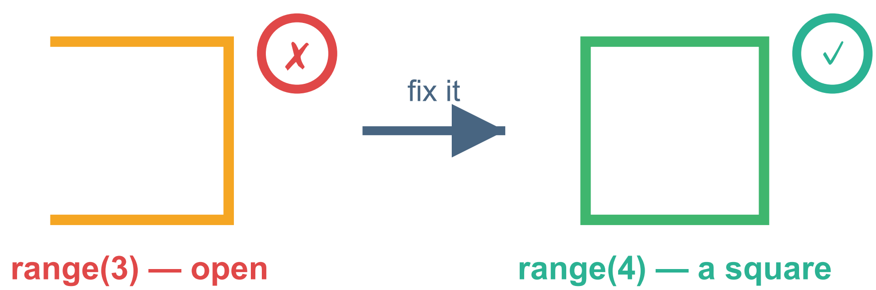
Bit should draw a SQUARE — but this is buggy

PYTHON

```python
for i in range(3):
    bit.forward(50)
    bit.right(90)
```

OUTPUT

```
Only 3 sides — the square does not close.
```



range(3) leaves the square open — fix it to range(4).

1 Find the bug

Bit should draw a square, but the code says range(3). What range number fixes it? range(______)

____

2 Fix the turn

Bit should draw a TRIANGLE, but the code is: for i in range(3): bit.forward(50); bit.right(90). The turn is wrong. What should right() be? (Hint: a triangle turns 120.)

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3 Fix the count

Bit should draw a straight line 100 long, moving 20 each time. The code says range(3). What range number fixes it? range(______)

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4 Challenge

Bit draws a square with range(4), forward(50). To make a BIGGER square, a student changed range(4) to range(8). That did NOT work. What should they change instead, and why?

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Big idea

To debug turtle code: say what it SHOULD draw, then check the range (how many sides) and the turn. Change the one that is wrong — to resize, change forward, not the range.

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Fix the Turtle Code

Quick facilitation notes & answer key

What this worksheet teaches

Students debug drawing instructions: decide what Bit should draw, then find and fix the wrong number — either how many times it repeats or the turn. The challenge shows that making a shape BIGGER means changing the move distance, not the repeat number. No coding experience needed.

Before you start — showing the drawing (optional)

The worksheet shows the typed instructions AND a picture of what Bit should draw, so you can teach straight from the printout — no computer required.

To show it live (optional): open a browser, go to a free "Python turtle" playground (e.g. search "trinket python turtle"), type the few lines from the worksheet, and press Run — a turtle draws it. You only press Run.

How to lead it (about 30 minutes)

1. Set the routine: "Decide what it SHOULD draw, run it, find the wrong number, fix it, run again."
2. Show it (you do). Take the square-that-should-be: it repeats 3, but a square needs 4 — fix the repeat number and run.
3. Let them work. Students do Exercises 1–4 (Ex 4 is the stretch), solo or in pairs.
4. Wrap up: the repeat number sets the number of sides; the move distance sets the size — different jobs.

Answer key (these are the verified correct answers)

Exercise 1 — repeat 4 times.

Exercise 2 — turn right 120 (a triangle turns 120).

Exercise 3 — repeat 5 times (5 × 20 = 100).

Exercise 4 (challenge) — change the move forward to a bigger number and keep repeating 4 times; repeating 8 times just retraces the same square, while the move distance sets the size.

If students get stuck — and ways to adjust

Watch for: changing the repeat number to resize a shape — that adds sides instead. Size comes from the move distance.

Make it easier: tell students what the shape should be, and have them fix just one number.

Make it harder: ask how to make a triangle bigger (increase the move forward, keep repeating 3 times).

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

In plain terms: students write and run step-by-step instructions — including a "repeat" instruction — to make something happen.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions, change one, and explain what the change did to the result.

You'll know it worked when

The student fixes the wrong repeat count or turn (Ex 1–3) and explains that size comes from the move distance, not the repeat number (Ex 4).